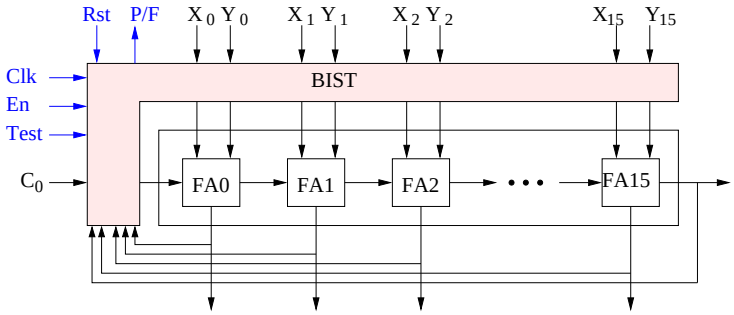
EE6083 VLSI Testing—Homework 1

1. Design a built-in self-test (BIST) circuit for an 16-bit ripple carry adder using Verilog. The BIST should be able to generate test patterns to cover 100% single cell faults. Consider the figure shown below, the additional test pins are in blue. Test is used to switch the circuit between normal mode and test mode. En is used to assert the embedded pattern generator and P/F is used to indicate the ripple carry adder is fault-free or faulty. Verify your BIST design by using fault-injection mechanism. (5 points)



2. Built-in self-repair (BISR) technique is widely used to improve the yield of regular circuits. Modify the 16-bit ripple carry adder by adding a spare full adder such that it can be repaired if one of the full adders in the ripple carry adder is faulty. Then, design a BISR circuit for the repairable ripple carry adder. Verify your BISR design by using fault-injection mechanism. (5 points)